

DATA LINK LAYER

Module 2

Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA) Protocol:

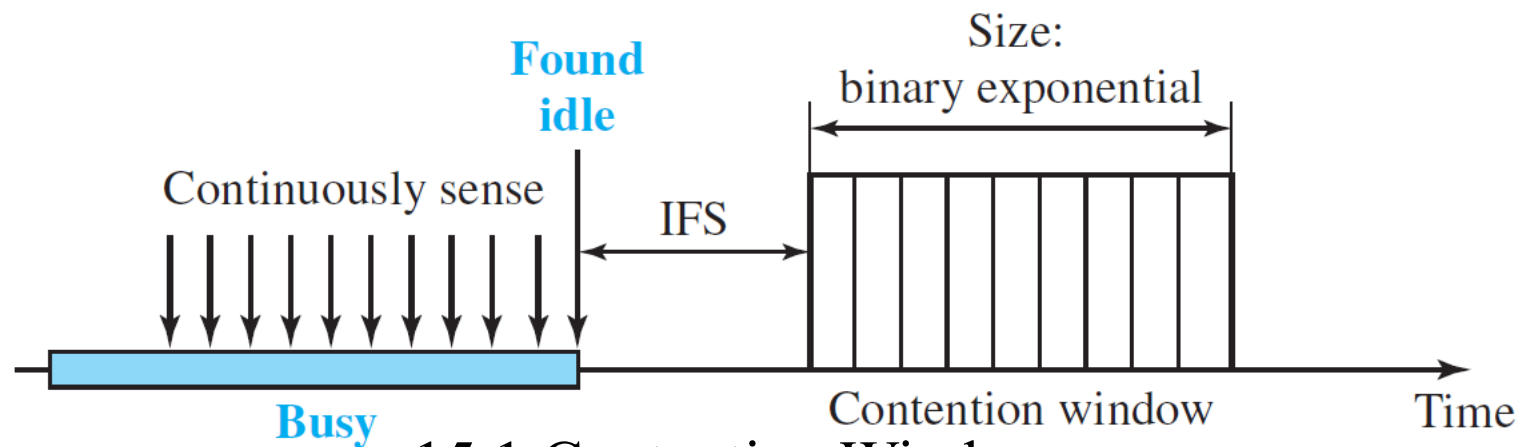
- Carrier sense multiple access with collision avoidance (**CSMA/CA**) was **invented** for **wireless networks**.
- **Collisions** are **avoided** through the use of CSMA/CA's **three strategies**: the **interframe space**, the **contention window**, and **acknowledgments**.

Interframe Space (IFS):

- In CSMA/CA, a station **does not transmit immediately** even if the **channel is found idle**.
- It **first waits** for a **short period** called the **Interframe Space (IFS)**. This **waiting time helps avoid collisions** because a **distant station** may have **already started transmitting**, but its **signal has not yet reached** the station.
- The IFS gives the **signal enough time to reach the station**. If the **channel is still idle** after the **IFS**, the station **does not transmit immediately**. It **waits** for a **random backoff time** based on the **contention window**.
- A **shorter IFS** means **higher priority**, so the station can **access the channel sooner**.

Contention Window:

- The **contention window** is a **period of time** divided into **time slots**. A **station** that **wants to transmit** randomly selects the **number of slots** it will **wait**. The **size** of the **contention window** increases using **binary exponential backoff**. It **starts** with **one slot** and **doubles** when the station **cannot transmit**.
- After each **time slot**, the station **senses the channel**. If the channel is **idle**, the **timer continues**. If the channel becomes **busy**, the station **pauses the timer** and **resumes it** when the **channel becomes idle again**. This **random waiting time** helps reduce the **chance of collisions**.



15.1 Contention Window

Acknowledgment:

- Even with these precautions, a collision can still occur, causing the frame to be lost. The data can also be corrupted during transmission. A positive acknowledgment (ACK) and a timeout timer help confirm that the receiver has successfully received the frame.

Legend

K : Number of attempts
 T_B : Backoff time
 IFS: Interframe Space
 RTS: Request to send
 CTS: Clear to send

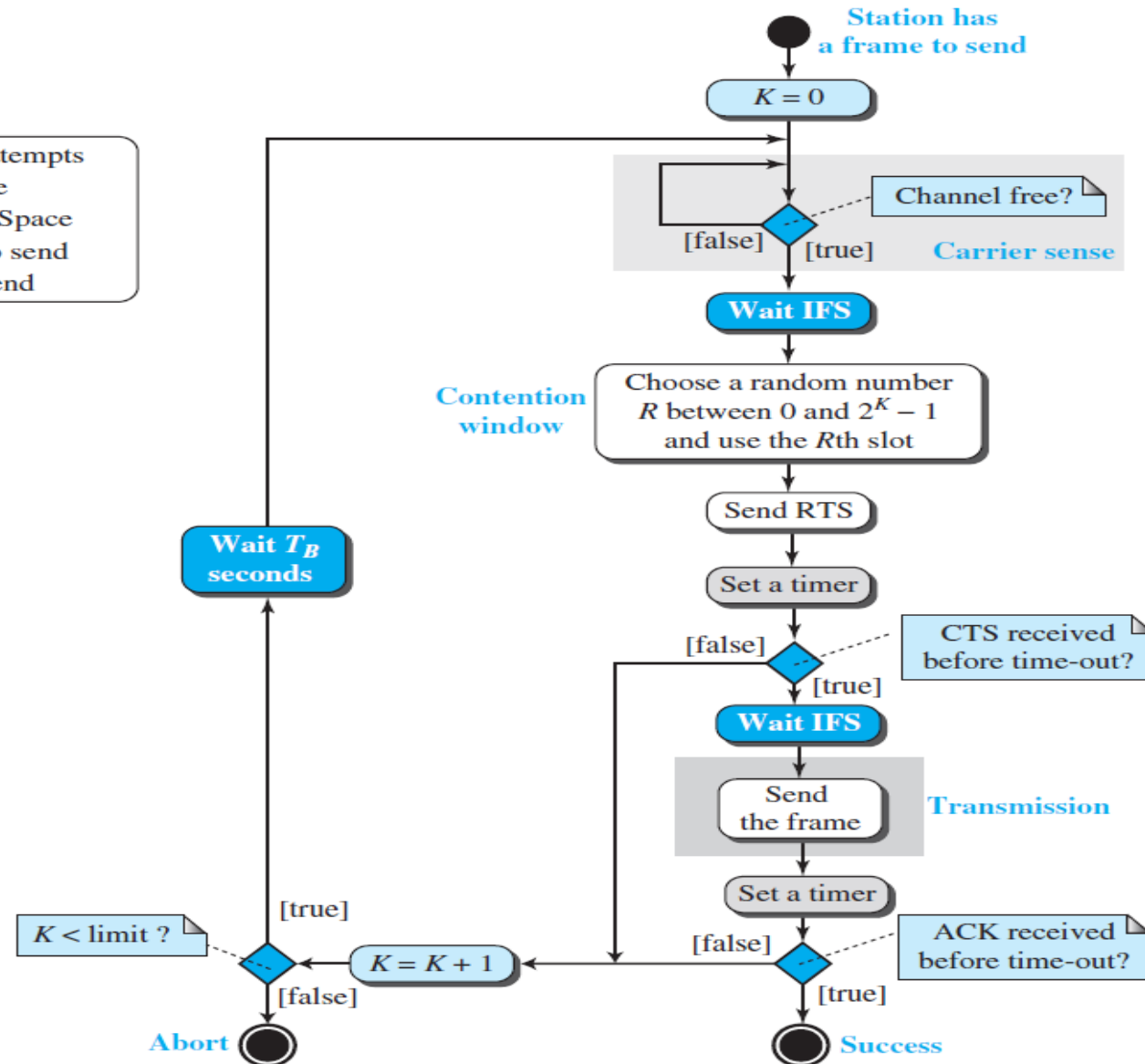


Fig. 15.2 Flow Diagram of CSMA/CA

CSMA/CA Frame Exchange Example:

- **Sense the channel:** Before sending, the source station checks whether the channel is idle. If the channel is busy, it waits using the backoff procedure.

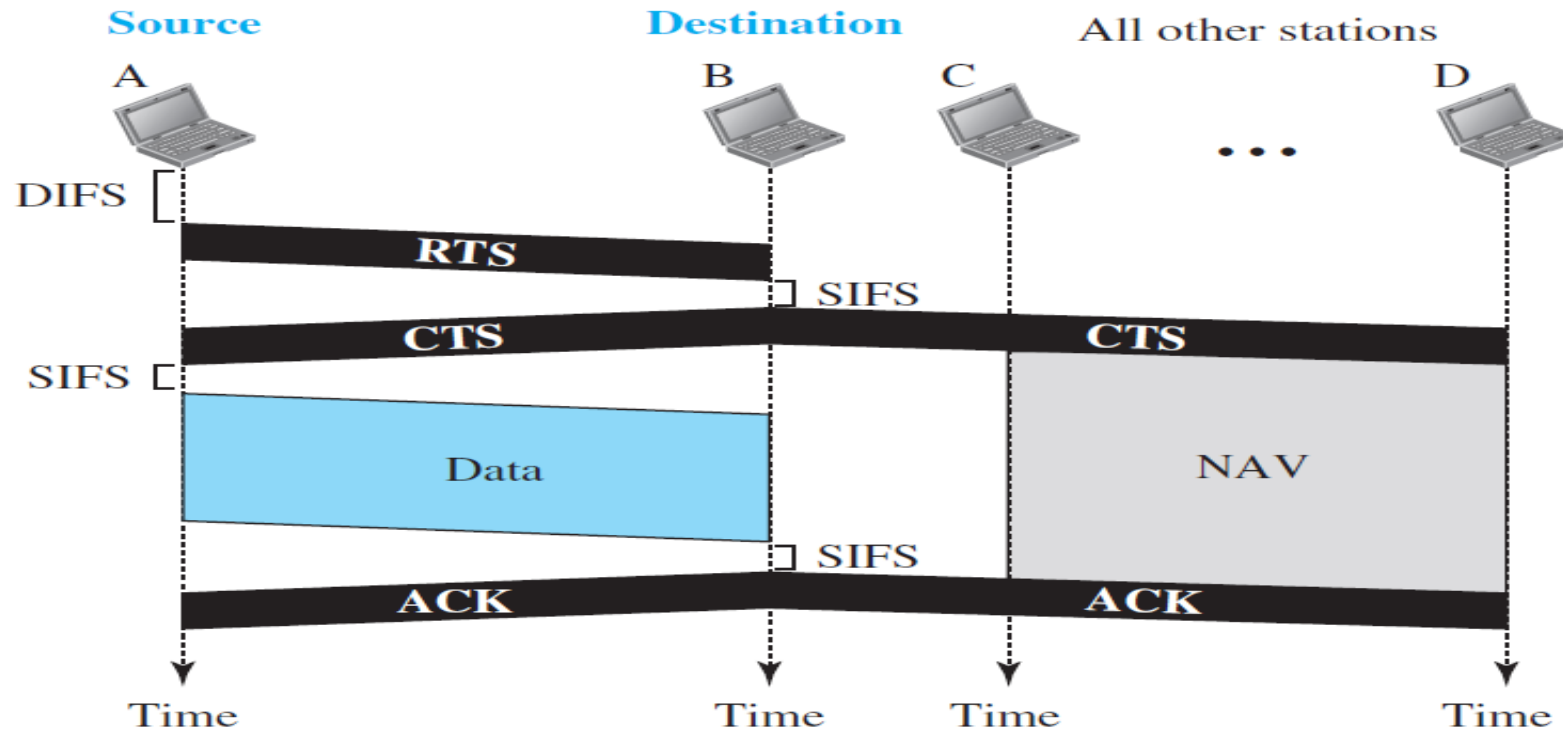


Fig 15.3 Frame Exchange of CSMA/CA

CSMA/CA Frame Exchange Example:

- **Send RTS:** When the channel becomes **idle**, the **source waits** for a **DIFS (DCF Interframe Space)** and then **sends an RTS (Request to Send)** frame to **the destination**.
- **Send CTS:** The **destination receives the RTS**, **waits** for a **SIFS (Short Interframe Space)**, and sends a **CTS (Clear to Send)** frame. The **CTS indicates** that the **destination is ready to receive the data**.
- **Send data:** **After receiving the CTS**, the **source waits for SIFS** and then **sends the data frame**.
- **Send ACK:** **After receiving the data**, the **destination waits for SIFS** and **sends an ACK (Acknowledgment)** to confirm that the **data was received successfully**.
- **Other stations wait:** **Other stations that hear the RTS or CTS** use **NAV (Network Allocation Vector)** to **know how long the channel will be busy**. They **remain silent during this period**.

Network Allocation Vector (NAV):

- NAV is used in CSMA/CA to help avoid collisions.
- When a station sends an RTS, it includes the amount of time it needs to use the channel.
- Other stations that hear the RTS set a NAV timer for this duration.
- While the NAV timer is running, these stations do not try to access the channel.
- Before checking whether the physical channel is idle, a station first checks its NAV.
- If the NAV has not expired, the station waits.
- If the NAV has expired, the station can check the channel and try to transmit.

Collision During Handshaking:

- A collision can occur when two or more stations send RTS frames at the same time.
- In CSMA/CA, the station cannot directly detect the collision.
- If the sender does not receive a CTS (Clear to Send) frame, it assumes that a collision has occurred.
- The station then uses the backoff procedure and waits for a random time.
- After the backoff period, the station tries to send the RTS again.

Hidden-Station Problem:

- The **RTS and CTS handshake** is used to solve the hidden station problem.
- The source station **B** sends an **RTS** to **A** in Fig. 15.3. The **RTS** reaches **A** but does not reach the hidden station **C**.
- **A** sends a **CTS** to **B**, and the **CTS** also reaches **C** because **C** is within **A**'s range.
- The **CTS** contains the **duration of the data transmission**.
- When **C** receives the **CTS**, it **knows** that **another station** is **using the channel**.
- Therefore, **C** **waits** until the **specified time is over** before **trying to transmit**.

Controlled Access Protocol:

- In controlled access, the stations **consult one another** to find **which station has the right to send**.
- A station **cannot send** unless it has been **authorized by other stations**.

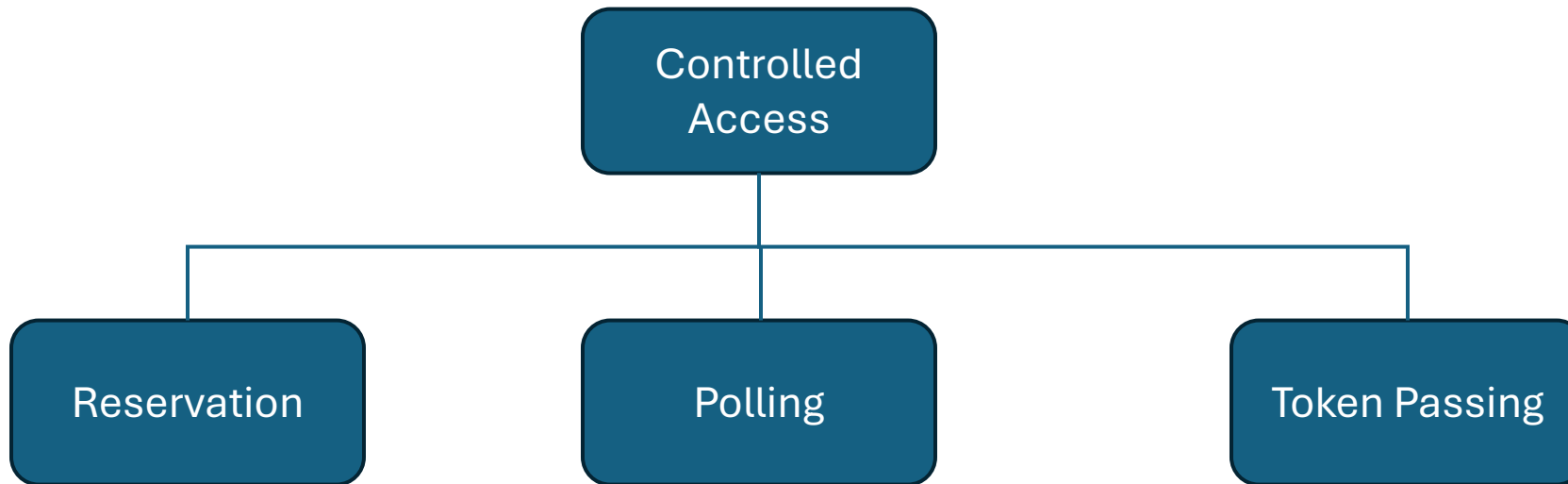


Fig 15.4 Types of Controlled Access Protocol

Reservation Protocol:

- In the **reservation method**, a station needs to **make a reservation before sending data**. **Time** is divided into **intervals**. In **each interval**, a **reservation frame** precedes the **data frames sent in that interval**.
- If there are **N stations** in the system, there are **exactly N reservation minislots** in the **reservation frame**. Each **minislot** belongs to a **station**. When a station needs to **send a data frame**, it **makes a reservation in its own minislot**.
- The **stations** that have **made reservations** can **send their data frames** after the **reservation frame**.

- Figure shows a situation with **five stations** and a **five-minislot reservation frame**. In the **first interval**, only **stations 1, 3, and 4** have made reservations.
- In the **second interval**, only **station 1** has made a reservation.

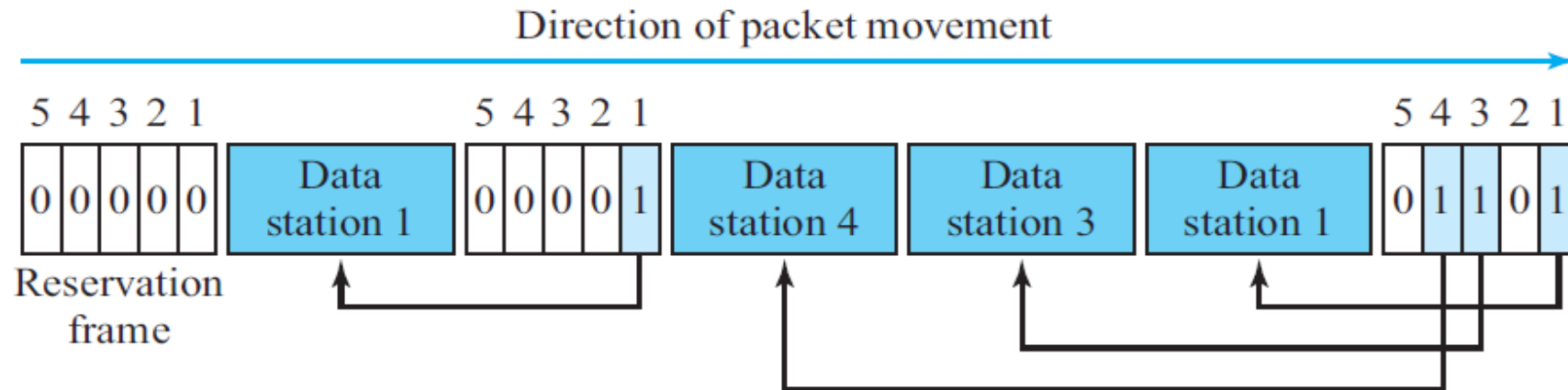


Fig 15.5 Reservation access method

Polling Protocol:

- In the polling method, **one device** is called the **primary station**, and **all other devices** are called **secondary stations**. **All communication** is **controlled** by the **primary station**, and **data exchanges** take place **through it**, even when the **sender and receiver** are **secondary stations**.
- The primary station **controls the channel** and **decides** which **secondary station** can use it. Therefore, the primary station **always starts** the **communication**. **Polling** uses two functions, **Select and Poll**, to **avoid collisions**.

Select Function:

- The Select function is used when the **primary station** wants to **send data to a secondary station**. The primary first **sends a SEL (Select) frame** containing the **address** of the **intended secondary station**.

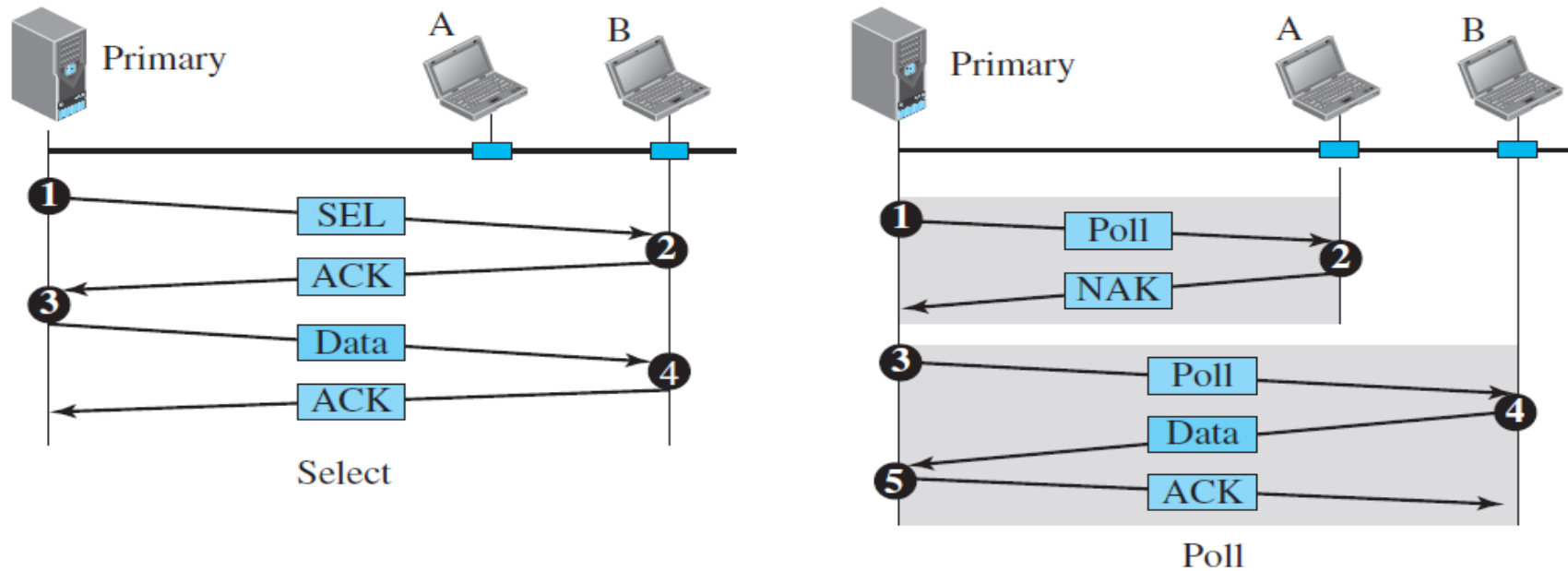


Fig 15.6 Select and poll functions in polling-access method

- This tells the secondary station that the **primary wants to send data**. The primary then **waits for an acknowledgment** that the **secondary is ready to receive**. After receiving the acknowledgment, the **primary sends the data**.

Poll Function:

- The Poll function is used when the primary station wants to **receive data** from **secondary stations**. The primary asks each **secondary station**, one by one, **whether it has data to send**. If a secondary has **no data**, it sends a **NAK response**.
- The primary then **polls the next station**. If a secondary has data, it **sends a data frame** to the primary. The primary **receives the data** and **sends an ACK** to confirm **successful receipt**.

Token Passing Protocol:

- In the token-passing method, **all stations** in the network are arranged in a **logical ring**. **Each station** has a **predecessor**, which is the station before it, and a **successor**, which is the station after it.
- A special packet called a **token continuously circulates** around the **ring**. **Possession** of the **token** gives a station the **right to access the channel** and **send data**. When a station has **data to send**, it **waits until** it **receives the token** from its **predecessor**. It then **holds** the **token** and **transmits its data**.
- **After** the station **finishes sending its data**, it **releases the token** and **passes it** to its **successor**. If a station **receives the token** but has **no data to send**, it simply **passes** the **token** to the **next station**. A station must **wait** for the **token** to **come around again** before it can **send more data**.

- Token management is important for the proper working of this method. It limits the amount of time a station can hold the token and ensures that the token is not lost or destroyed.
- For example, if a station holding the token fails, the token may disappear from the network. Token management also helps assign priorities to stations and to the types of data being transmitted. It can make a low-priority station release the token when a high-priority station needs access to the channel.

Channelization Protocol:

- Channelization (or channel partition) is a **multiple-access method** in which the **available bandwidth** of a link is **shared in time, frequency, or through code**, among different stations.

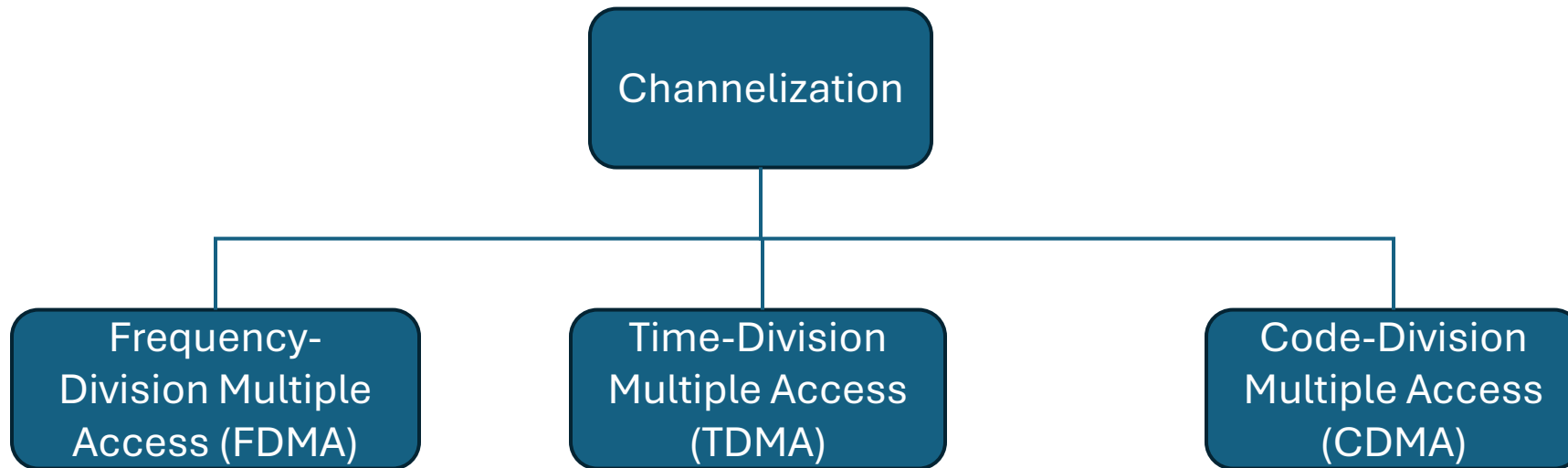


Fig 15.7 Types of Channelization Protocol



THANK YOU